

Enterprise RAG System Architecture & Evaluation Benchmark

Document Version: 2.1 | Deployment Pipeline Specification

1. System Overview

This specification document outlines the production architecture for the Automated RAG Evaluation and Quality Gate Service. The system provides high-throughput document retrieval, cross-encoder reranking, and automated CI/CD validation for large language model applications.

2. Technical Stack & Components

The core pipeline consists of the following key frameworks and infrastructure:

Component	Technology	Primary Purpose
Vector Database	ChromaDB	Dense embedding storage and similarity search
Sparse Search	BM25 (rank-bm25)	Exact keyword matching and token scoring
Fusion Engine	Reciprocal Rank Fusion	Blending dense and sparse retrieval ranks
Reranker Model	FlashRank Cross-Encoder	High-precision top-N context re-scoring
LLM Provider	Groq Cloud API	Ultra-low latency inference using Llama models
CI/CD Engine	GitHub Actions	Automated quality gate testing on code push

3. Operational Requirements & Quality Gates

To guarantee zero regression, all pull requests must pass an automated CI Quality Gate. The automated judge measures Faithfulness and Relevancy against a minimum threshold of 0.80. System deployments are automatically aborted if evaluation scores fall below this target threshold.

4. Guardrails & Refusal Policies

The RAG system enforces strict out-of-domain guardrails. When queries fall outside the context of the active document knowledge base, the system strictly outputs standard refusal messages: 'I cannot answer this based on the available documentation.' This prevents false hallucinations in mission-critical environments.

5. Performance Benchmarks & SLA Targets

Expected system latencies across production deployment stages:

Pipeline Stage	Target SLA	Model / Method Used
Hybrid Retrieval	< 50 ms	ChromaDB + BM25Okapi
Cross-Encoder Rerank	< 100 ms	ms-marco-MiniLM-L-12-v2
LLM Generation	< 1200 ms	Groq llama-3.1-8b-instant
Total End-to-End SLA	< 1500 ms	Full Pipeline Execution